UNIQUENESS IN THE VALUATIVE CRITERION IMPLIES THAT THE DIAGONAL IS SEPARATED

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The translation of uniqueness in the valuative criterion into a condition on the category of dotted arrows. Used throughout the counted proof (source lines 9633-9650). Turns the hypothesis into a statement about morpisms of dotted arrows.
- The description of the diagonal as a morphism satisfying the corresponding valuative criterion. Invoked inside the counted proof (source lines 9651-9668). Reduces separatedness of the diagonal to the uniqueness property for the diagonal.
- The valuative criterion for separatedness of a morphism of algebraic stacks. Invoked at the end of the counted proof (source lines 9669-9680). Concludes separatedness from the established uniqueness property.

Source reference. Original definition definition-fill-in-diagram: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9061–9061. Navigation pointer only: the counted text refers to the Stacks item definition-fill-in-diagram, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-cat-dotted-arrows: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9115–9115. Navigation pointer only: the counted text refers to the Stacks item lemma-cat-dotted-arrows, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma lemma-existence-representable: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9409–9409. Navigation pointer only: the counted text refers to the Stacks item lemma-existence-representable, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-properties-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 344–344. Navigation pointer only: the counted text refers to the Stacks item lemma-properties-diagonal, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-characterize-proper: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 9176–9176. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-characterize-proper, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. If f is separated, then f satisfies the uniqueness part of the valuative criterion.*

Proof. Since f is separated we see that all categories of dotted arrows are setoids by Lemma lemma-cat-dotted-arrows. Consider a diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \xrightarrow{x} & \mathcal{X} \\ {\scriptstyle j}\downarrow & \nearrow & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(A) & \xrightarrow{y} & \mathcal{Y} \end{array}$$

and a 2-morphism $\gamma : y \circ j \to f \circ x$ as in Definition definition-fill-in-diagram. Consider two objects (a, α, β) and (a', β', α') of the category of dotted arrows. To finish the proof we have to show these objects are isomorphic. The isomorphism

$$f \circ a \xrightarrow{\beta^{-1}} y \xrightarrow{\beta'} f \circ a'$$

means that $(a, a', \beta' \circ \beta^{-1})$ is a morphism $\mathrm{Spec}(A) \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$. On the other hand, α and α' define a 2-arrow

$$(a, a', \beta' \circ \beta^{-1}) \circ j = (a \circ j, a' \circ j, (\beta' \star \mathrm{id}_j) \circ (\beta \star \mathrm{id}_j)^{-1}) \xrightarrow{(\alpha, \alpha')} (x, x, \mathrm{id}) = \Delta_f \circ x$$

Here we use that both (a, α, β) and (a', α', β') are dotted arrows with respect to γ. We obtain a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \xrightarrow{x} & \mathcal{X} \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle \Delta_f} \\ \mathrm{Spec}(A) & \xrightarrow{(a, a', \beta' \circ \beta^{-1})} & \mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \end{array}$$

with 2-commutativity witnessed by (α, α'). Now Δ_f is representable by algebraic spaces (Lemma lemma-properties-diagonal) and proper as f is separated. Hence

by Lemma lemma-existence-representable and the valuative criterion for properness for algebraic spaces (Morphisms of Spaces, Lemma spaces-morphisms-lemma-characterize-proper) we see that there exists a dotted arrow. Unwinding the construction, we see that this means (a, α, β) and (a', α', β') are isomorphic in the category of dotted arrows as desired. $\square$